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In [2]: import tkinter as tk
from tkinter import messagebox, simpledialog, ttk

class ContactManager:
    def __init__(self, root):
        self.root = root
        self.root.title("Contact Manager")
        self.contacts = {}

        # Setup UI
        self.setup_ui()

    def setup_ui(self):
        self.frame = ttk.Frame(self.root, padding="10")
        self.frame.pack(fill=tk.BOTH, expand=True)

        self.title_label = ttk.Label(self.frame, text="Contact Manager", font=("Helvetica", 16))
        self.title_label.pack(pady=10)

        # Add Contact Button
        self.add_button = ttk.Button(self.frame, text="Add Contact", command=self.add_contact)
        self.add_button.pack(pady=5)

        # View Contacts Button
        self.view_button = ttk.Button(self.frame, text="View Contacts", command=self.view_contacts)
        self.view_button.pack(pady=5)

        # Search Contact Button
        self.search_button = ttk.Button(self.frame, text="Search Contact", command=self.search_contact)
        self.search_button.pack(pady=5)

        # Update Contact Button
        self.update_button = ttk.Button(self.frame, text="Update Contact", command=self.update_contact)
        self.update_button.pack(pady=5)

        # Delete Contact Button
        self.delete_button = ttk.Button(self.frame, text="Delete Contact", command=self.delete_contact)
        self.delete_button.pack(pady=5)

        # Exit Button
        self.exit_button = ttk.Button(self.frame, text="Exit", command=self.root.destroy)
        self.exit_button.pack(pady=5)
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def add_contact(self):
    name = simpledialog.askstring("Add Contact", "Enter contact name:")
    if name:
        while True:
            phone = simpledialog.askstring("Add Contact", "Enter contact phone (10 digits):")
            if phone and phone.isdigit() and len(phone) == 10:
                break
            else:
                messagebox.showwarning("Input Error", "Phone number must be exactly 10 digits.")
        email = simpledialog.askstring("Add Contact", "Enter contact email:")
        if email:
            address = simpledialog.askstring("Add Contact", "Enter contact address:")
            if address:
                if name in self.contacts:
                    messagebox.showwarning("Duplicate Error", "Contact already exists.")
                else:
                    self.contacts[name] = {"phone": phone, "email": email, "address": address}
                    messagebox.showinfo("Success", "Contact added successfully!")
            else:
                messagebox.showwarning("Input Error", "Address is required.")
        else:
            messagebox.showwarning("Input Error", "Email is required.")
    else:
        messagebox.showwarning("Input Error", "Name is required.")

def view_contacts(self):
    contacts_str = "\n".join([f"Contact {i+1}:\nName: {name}\nPhone: {details['phone']}\nEmail: {details['email']}\nAddress: {details['address']}" for i, (name, details) in enumerate(self.contacts.items())])
    messagebox.showinfo("Contact List", contacts_str if contacts_str else "No contacts found.")

def search_contact(self):
    query = simpledialog.askstring("Search Contact", "Enter name or phone number to search:")
    if query:
        results = [f"Contact {i+1}:\nName: {name}\nPhone: {details['phone']}\nEmail: {details['email']}\nAddress: {details['address']}" for i, (name, details) in enumerate(self.contacts.items()) if query in name or query in details['phone']]
        messagebox.showinfo("Search Results", "\n".join(results) if results else "No contacts found.")

def update_contact(self):
    name = simpledialog.askstring("Update Contact", "Enter the name of the contact to update:")
    if name in self.contacts:
        current_details = self.contacts[name]
        messagebox.showinfo("Current Details", f"Phone: {current_details['phone']}\nEmail: {current_details['email']}\nAddress: {current_details['address']}")
        while True:
            phone = simpledialog.askstring("Update Contact", f"Enter new phone number (current: {current_details['phone']}):")
            if phone and phone.isdigit() and len(phone) == 10:
                break
            else:
                messagebox.showwarning("Input Error", "Phone number must be exactly 10 digits.")
        current_details['phone'] = phone
    else:
        messagebox.showwarning("Input Error", "Contact not found.")

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        if phone and phone.isdigit() and len(phone) == 10:
            break
        else:
            messagebox.showwarning("Input Error", "Phone number must be exactly 10 digits.")

    email = simplifiedialog.askstring("Update Contact", f"Enter new email (current: {current_details['email']}):", initialv
    if email:
        address = simplifiedialog.askstring("Update Contact", f"Enter new address (current: {current_details['address']}):",
        if address:
            self.contacts[name] = {"phone": phone, "email": email, "address": address}
            messagebox.showinfo("Success", "Contact updated successfully!")
        else:
            messagebox.showwarning("Input Error", "Address is required.")
    else:
        messagebox.showwarning("Input Error", "Email is required.")
else:
    messagebox.showwarning("Error", "Contact not found.")

def delete_contact(self):
    name = simplifiedialog.askstring("Delete Contact", "Enter the name of the contact to delete:")
    if name in self.contacts:
        del self.contacts[name]
        messagebox.showinfo("Success", "Contact deleted successfully!")
    else:
        messagebox.showwarning("Error", "Contact not found.")

if __name__ == "__main__":
    root = tk.Tk()
    app = ContactManager(root)
    root.mainloop()

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In []: